求 C 和 b_c 的梯度：

先考虑单样本 记 $\delta_j^{(i)} = \frac{\partial l^{(i)}}{\partial z_j} = \hat{y}_j^{(i)} - y_j^{(i)}$

考虑 C 中的一个元素 C_{ab} $\frac{\partial z_k}{\partial C_{ab}} = \begin{cases} r_b, & k=a \\ 0, & k\neq a \end{cases}$

由链式法则 $\frac{\partial l}{\partial C_{ab}} = \sum_{k=1}^{K} \frac{\partial l}{\partial z_k} \frac{\partial z_k}{\partial C_{ab}} = \delta_a r_b$

因此 $\nabla_C l = \begin{bmatrix} \delta_1 r_1 & \cdots & \delta_1 r_D \\ \vdots & & \vdots \\ \delta_K r_1 & \cdots & \delta_K r_D \end{bmatrix}$

$= \vec{\delta}\,\vec{r}^{\,T}$

考虑 $\vec{b}_c$ 中的一个元素 $b_{c,a}$ $\frac{\partial z_k}{\partial b_{c,a}} = \begin{cases} 1, & k=a \\ 0, & k\neq a \end{cases}$

由链式法则 $\frac{\partial l}{\partial b_{c,a}} = \sum_{k=1}^{K} \frac{\partial l}{\partial z_k} \frac{\partial z_k}{\partial b_{c,a}} = \delta_a$

因此 $\nabla_{\vec{b}_c} l = \vec{\hat{y}} - \vec{y} = \vec{\delta}$

推广到小批量平均可得

$$\nabla_C L = \frac{1}{m} \sum_{n=1}^{m} \vec{\delta}^{(n)} \vec{r}^{(n)T} = \frac{1}{m} \sum \begin{bmatrix} \delta_1^{(n)} r_1^{(n)} & \cdots & \delta_1^{(n)} r_D^{(n)} \\ \vdots & & \\ \delta_K^{(n)} r_1^{(n)} & \cdots & \delta_K^{(n)} r_D^{(n)} \end{bmatrix}$$

$$\nabla_{\vec{b}_c} L = \frac{1}{m} \sum_{n=1}^{m} \left(\vec{\hat{y}}^{(n)} - \vec{y}^{(n)} \right) = \frac{1}{m} \sum_{n=1}^{m} \vec{\delta}^{(n)}$$

求 $W_{2,e}$ 和 $b_{2,e}$ 的梯度

先考虑单个样本

考虑 $\vec{r}$ 中的一个元素 r_d $\quad \frac{\partial z_k}{\partial r_d} = C_{kd}$

由链式法则 $\quad \frac{\partial l}{\partial r_d} = \sum_{k=1}^{K} \frac{\partial l}{\partial z_k} \frac{\partial z_k}{\partial r_d} = \sum_{k=1}^{K} \delta_k C_{kd}$

记 $\quad \vec{a} = \nabla_{\vec{r}} l = C^T \vec{\delta}$

考虑 $\vec{f}_e$ 的一个分量 $f_{e,d}$ 有 $\quad \frac{\partial r_t}{\partial f_{e,d}} = \begin{cases} g_e, & t=d \\ 0, & t \neq d \end{cases}$

由链式法则 $\quad \frac{\partial l}{\partial f_{e,d}} = \sum_{t=1}^{D} \frac{\partial l}{\partial r_t} \frac{\partial r_t}{\partial f_{e,d}} = a_d g_e$

所以 $\quad \nabla_{f_e} l = g_e C^T \vec{\delta}$

考虑 $W_{2,e}$ 的一个元素 $(W_{2,e})_{ab}$ $\quad \frac{\partial f_{e,d}}{\partial (W_{2,e})_{ab}} = \begin{cases} u_{e,b}, & d=a \\ 0, & d \neq a \end{cases}$

由链式法则 $\quad \frac{\partial l}{\partial (W_{2,e})_{ab}} = \sum_{d=1}^{D} \frac{\partial l}{\partial f_{e,d}} \frac{\partial f_{e,d}}{\partial (W_{2,e})_{ab}} = g_e a_a u_{e,b}$

所以 $\quad \nabla_{W_{2,e}} l = (g_e \vec{a}) \vec{u}_e^T$

考虑 $\vec{b}_{2,e}$ 的一个分量 $\quad \frac{\partial f_{e,d}}{\partial (b_{2,e})_a} = \begin{cases} 1, & d=a \\ 0, & d \neq a \end{cases}$

由链式法则 $\quad \frac{\partial l}{\partial (b_{2,e})_a} = \sum_{d=1}^{D} \frac{\partial l}{\partial f_{e,d}} \frac{\partial f_{e,d}}{\partial (b_{2,e})_a} = g_e a_a$

所以 ~~$\nabla_{b_{2,e}}$~~ $\nabla_{b_{2,e}} l = g_e \vec{a}$

加入小批量平均

$$\nabla_{W_{2,e}} L = \frac{1}{m} \sum_{n=1}^{m} g_e^{(n)} \vec{a}^{(n)} (\vec{u}_e^{(n)})^T$$

$$\nabla_{b_{2,e}} L = \frac{1}{m} \sum_{n=1}^{m} g_e^{(n)} \vec{a}^{(n)}$$

求 $W_{1,e}$ 和 $b_{1,e}$ 的梯度

记 $\bar{r} = \nabla_r l \quad \bar{f}_e = \nabla_{f_e} l$

先求 $\nabla_{u_e} l$：$\frac{\partial f_{e,d}}{\partial u_{e,q}} = (W_{2,e})_{dq}$

所以 $$\frac{\partial l}{\partial u_{e,q}} = \sum_{d=1}^{D} \frac{\partial l}{\partial f_{e,d}} \cdot \frac{\partial f_{e,d}}{\partial u_{e,q}}$$

$$= \sum_{d=1}^{D} (\bar{f}_e)_d (W_{2,e})_{dq}$$

记 $\bar{u}_e = \nabla_{u_e} l = g_e W_{2,e}^T \bar{r}$

再求 $\nabla_{v_e} l$：$$\frac{\partial u_{e,q}}{\partial v_{e,q}} = \begin{cases} 1, & v_{e,q} > 0 \\ 0, & v_{e,q} \le 0 \end{cases}$$

所以 $$\frac{\partial l}{\partial v_{e,q}} = \frac{\partial l}{\partial u_{e,q}} 1_{\{v_{e,q} > 0\}}$$

记 $M_e = 1_{\{v_e > 0\}}$

则 $\bar{v}_e = \nabla_{v_e} l = \bar{u}_e \odot M_e = (g_e W_{2,e}^T \bar{r}) \odot 1_{\{v_e > 0\}}$

再对 $W_{1,e}$ 求导

$$\frac{\partial v_{e,q}}{\partial (W_{1,e})_{ab}} = \begin{cases} h_b, & q=a \\ 0, & q\neq a \end{cases}$$

$$\frac{\partial \ell}{\partial (W_{1,e})_{ab}} = \sum_{q=1}^{H} \frac{\partial \ell}{\partial v_{e,q}} \cdot \frac{\partial v_{e,q}}{\partial (W_{1,e})_{ab}}$$

$$= (\bar{v}_e)_a h_b$$

所以 $\nabla_{W_{1,e}} \ell = \bar{v}_e h^T$

再对 $b_{1,e}$ 求导

$$\frac{\partial v_{e,q}}{\partial (b_{1,e})_a} = \begin{cases} 1, & q=a \\ 0, & q\neq a \end{cases}$$

$$\frac{\partial \ell}{\partial (b_{1,e})_a} = \frac{\partial \ell}{\partial v_{e,a}} = (\bar{v}_e)_a$$

所以 $\nabla_{b_{1,e}} \ell = \bar{v}_e$

加入小批量平均

$$\nabla_{W_{1,e}} L = \frac{1}{m} \sum_{n=1}^{m} \bar{v}_e^{(n)} (h^{(n)})^T$$

$$\nabla_{b_{1,e}} L = \frac{1}{m} \sum_{n=1}^{m} \bar{v}_e^{(n)}$$

求G的梯度

记 $\rho = Gh \quad g = \text{softmax}(\rho)$

先求g的梯度

$$\frac{\partial r_d}{\partial g_e} = f_{e,d}$$

$$\frac{\partial l}{\partial g_e} = \sum_{d=1}^{D} \frac{\partial l}{\partial r_d} \frac{\partial r_d}{\partial g_e} = f_e^T \bar{r}$$

记 $\beta_e = \frac{\partial l}{\partial g_e} = f_e^T \bar{r}$

$$\beta = [\beta_1, \cdots, \beta_E]^T$$

再求ρ的梯度

$$\frac{\partial g_j}{\partial \rho_e} = g_j (1_{\{j=e\}} - g_e)$$

$$\frac{\partial l}{\partial \rho_e} = \sum_{j=1}^{E} \frac{\partial l}{\partial g_j} \frac{\partial g_j}{\partial \rho_e} = \beta_e g_e - g_e \sum_{j=1}^{E} g_j \beta_j$$

记 $\bar{\rho}_e = \frac{\partial l}{\partial \rho_e} = \beta_e g_e - g_e \sum_{j=1}^{E} g_j \beta_j$

再求G的梯度

$$\frac{\partial \rho_e}{\partial G_{ab}} = \begin{cases} h_b, & e = a \\ 0, & e \neq a \end{cases}$$

$$\frac{\partial l}{\partial G_{ab}} = \sum_{e=1}^{E} \frac{\partial l}{\partial \rho_e} \frac{\partial \rho_e}{\partial G_{ab}} = \bar{\rho}_a h_b$$

所以 $\nabla_G \ell = \bar{p} h^T$

加入小批量平均

$$\nabla_G L = \frac{1}{m} \sum_{n=1}^{m} \bar{p}^{(n)} (h^{(n)})^T$$

求 h 的梯度

残差路径：$\left.\frac{\partial r}{\partial h}\right|_{res} = I_D$

$$\nabla_h \ell|_{res} = \bar{r}$$

专家路径：$\left.\frac{\partial v_{e,q}}{\partial h_d}\right|_{exp} = (W_{1,e})_{qd}$

$$\left.\frac{\partial \ell}{\partial h_d}\right|_{exp\,e} = (W_{1,e}^T \bar{v}_e)_d$$

$$\nabla_h \ell|_{exp} = \sum_{e=1}^{E} W_{1,e}^T \bar{v}_e$$

路由器路径：$\frac{\partial p_e}{\partial h_d} = G_{e,d}$

$$\left.\frac{\partial \ell}{\partial h_d}\right|_{rout} = \sum_{e=1}^{E} \bar{p}_e G_{ed}$$

$$\nabla_h \ell|_{rout} = G^T \bar{p}$$

所以 $\nabla_h \ell = \bar{r} + \sum_{e=1}^{E} W_{1,e}^T \bar{v}_e + G^T \bar{p}$

加入小批量平均 $\nabla_{W_1} L = \frac{1}{m} \sum_{n=1}^{m} \left[\bar{r}^{(n)} + \sum_{e=1}^{E} W_{1,e}^T \bar{v}_e^{(n)} + G^T \bar{p}^{(n)} \right]$

数值计算 $\frac{\partial L}{\partial G_{1,1}}$：

$$\bar{r} = C^T(\hat{y} - y) = \begin{bmatrix} 1/5 & 0 & 1/5 \\ 0 & 1/5 & 1/5 \end{bmatrix} \begin{bmatrix} 1/3 \\ -2/3 \\ 1/3 \end{bmatrix} = \begin{bmatrix} 2/15 \\ -1/15 \end{bmatrix}$$

$$\beta_1 = f_1^T \bar{r} = \frac{1}{50}$$

$$\beta_2 = f_2^T \bar{r} = \frac{1}{25}$$

$$\frac{\partial L}{\partial p_1} = g_1(\beta_1 - g_1\beta_1 - g_2\beta_2) = -\frac{1}{200}$$

$$\frac{\partial L}{\partial G_{1,1}} = \frac{\partial L}{\partial p_1} h_1 = -\frac{1}{100}$$